

The Indian Pharma Landscape: A First-Principles Primer

A first-principles guide to innovators, API, formulation and contract manufacturers.

6.7%

of drugs entering human trials reach approval (Citeline, 2014 to 2023)

7 to 14 yrs

of patent-protected selling time after launch: about 7 to 10 before extensions, up to 14 in the US with them

89%

of US prescriptions are generics, but they make up only 11.6% of total US prescription drug spending (2025)

\$300B+

of annual global sales of branded prescription drugs lose patent protection between 2025 and 2030

Evaluate is a pharma research firm (part of Norstell) that forecasts drug sales. Other estimates of the same 2025 to 2030 patent cliff: IQVIA, about \$220B of global sales affected as biologics face biosimilar competition; Morgan Stanley, about \$175B of US drug revenue at risk; trade-press estimates as high as \$400B.

● Pharma is three businesses under one label:

1. **Innovators** discover molecules and earn monopoly-style profits for a limited window.
2. **Generic makers** copy the molecule once the patent ends and compete on price.
3. **Contract manufacturers** make the molecule for whoever owns it and earn contract margins.

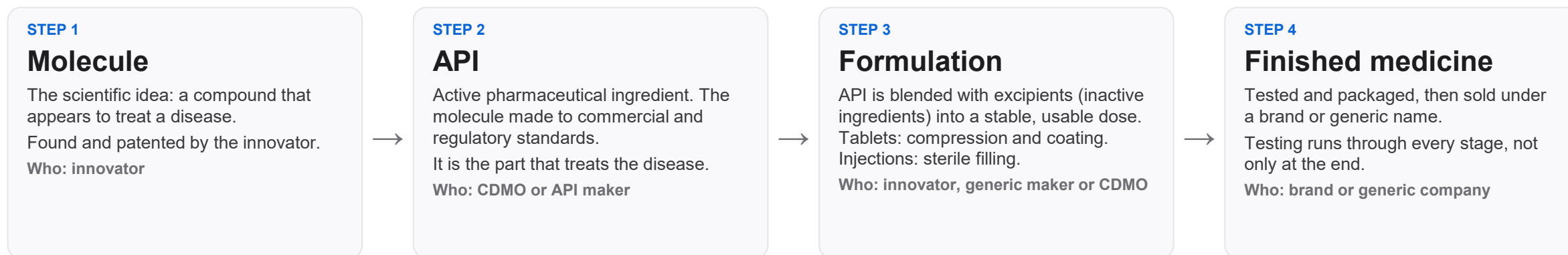
Most investment questions in the sector reduce to three things: which of the three a company is, how hard its product is to copy, and whether its plants stay in good standing with the US regulator.

Inside: the value chain · the patent clock and cliff · who does what (CRO, CDMO, CMO) · business models · therapeutic areas · complexity as a moat · India · the 2025 to 2030 patent cliff · semaglutide case study · FDA and risks · investor checklist

Sources include US FDA, IQVIA, Citeline, HHS/ASPE, the Association for Accessible Medicines, Novo Nordisk and Dr. Reddy's filings, IBEF and company disclosures; each slide lists its own. Some figures come from secondary sources and are marked as estimates.

A molecule becomes a medicine in four steps, and a different kind of company runs each one

Follow one drug from idea to pharmacy shelf.



● Why the API is not the medicine

The API is concentrated and too potent to take on its own.
Excipients add bulk, stability and a way to deliver a precise dose.
The result is what a patient actually takes.

● Why the kind of drug matters

How a drug is made decides how hard it is to copy, how many firms can make it, and what the copy is called. The GLP-1 obesity drugs and the cancer antibody Keytruda sit in very different boxes.

The same word, "molecule", covers all three types. Investors care about the type because it sets the barrier to entry.

Exhibit 1 Three kinds of drug

TYPE	HOW IT IS MADE	EXAMPLES	A COPY IS CALLED
Small molecule	Chemical synthesis in reactors	Most tablets e.g. apixaban (Eliquis)	Generic
Peptide	Amino acids linked step by step in a chain; harder than small molecules	Semaglutide, tirzepatide	Generic
Biologic	Large proteins grown in living cells	Keytruda, Stelara, insulin	Biosimilar

Source: company and FDA product information; classification simplified for a general reader.

Most of a patent's life is spent getting to market, so the innovator's window to earn is limited

Why prices start high, and why the end of a patent is such a big moment.

● The innovator's bet

An innovator discovers the molecule and owns the patent. Imagine a company that finds a molecule that kills a type of cancer cell. Getting there means screening and refining thousands of candidate compounds before one works, then proving it in human trials. About 6.7% of drugs entering human trials are approved.

Development is commonly put at 10 to 15 years, and cost estimates run from roughly \$0.7 billion (median R&D cost) to \$2.6 billion (Tufts estimate) per approved drug, depending on method.

● The clock

A patent lasts about 20 years from filing, not from launch. Development uses up 10 or more of those years, leaving roughly 7 to 10 years of protected sales.

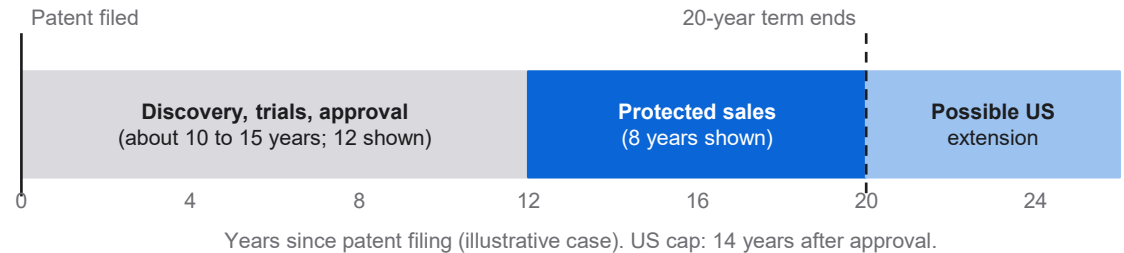
The innovator charges high prices in that window to recover R&D. In the US, a patent term extension of up to 5 years can add time, capped at 14 years after approval. The EU offers similar extensions.

● The cliff

When protection ends, any qualified generic maker can sell the same molecule. In the US, the first company to challenge a patent (a Paragraph IV filing) can earn 180 days of generic exclusivity.

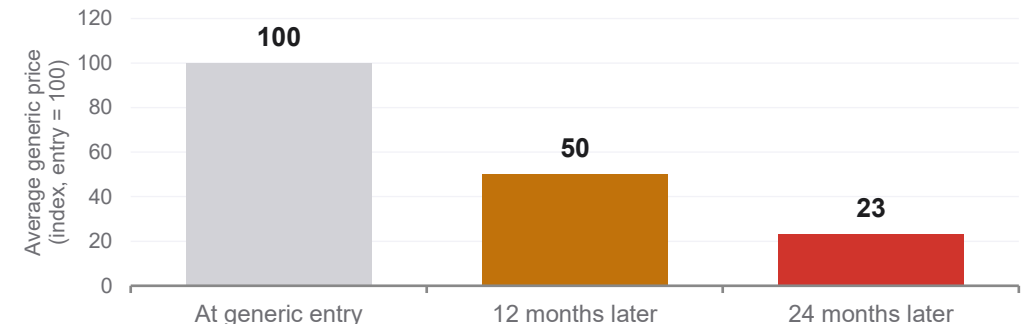
After that, more entrants arrive and prices fall fast: more than 10 generic makers are frequently competing within two years.

Exhibit 2 The patent clock, illustrated



Source: Illustrative timeline built from typical ranges; actual timing varies by drug and jurisdiction. US patent term extension rules: 35 U.S.C. 156 (up to 5 years, 14-year cap after approval).

Exhibit 3 Generic prices after entry



Source: HHS ASPE (2016): average generic prices fell by half in 12 months and by 77% in 24 months after entry; chart is an index built from those two figures. ASPE (2021): with 10 or more competitors, prices reach roughly 20 to 30% of the pre-entry brand price. Approval rate: Citeline/Norstell. Cost: RAND, JAMA Netw Open 2025; Tufts CSDD 2014.

CRO, CDO, CMO, CDMO and CRDMO describe layers of capability, not different industries

Read the acronyms as a stack: research, then development, then manufacturing. Companies differ in how many layers they cover.

Exhibit 4 Capability stack by company type

TYPE	RESEARCH (DISCOVERY, PRECLINICAL)	PROCESS DEVELOPMENT (ROUTE DESIGN, SCALE-UP)	API MANUFACTURING	FORMULATION AND FINISHED DOSE	OWNS THE PATENT	HOW IT EARNS	INDIAN EXAMPLES (INDICATIVE)
CRO Contract research org.	Core	Some	No	No	No	Fees per project or per scientist	Research arms of the CRDMOs below
CDO Contract development org.	No	Core	Some	Some	No	Fees for development work; no commercial-scale supply	Development arms of CDMOs
CMO Contract manufacturing org.	No	No	Core	Some	No	Per batch or volume, to the client's specification	Often a unit of a larger firm
CDMO Contract development and manufacturing org.	No	Core	Core	Some	No	Multi-year supply contracts; sometimes capacity advances	Divi's (custom synthesis), Cohance, Neuland
CRDMO Contract research, development and manufacturing org.	Core	Core	Core	Some	No	Fees from discovery through commercial supply	Syngene, Sai Life Sciences
Generic API maker Sells its own APIs	No	Some	Core	No	No	Sells API to formulators; price competition	Divi's and Laurus (generic segments)
Generic formulator Sells finished generics	No	Some	Some	Core	No	Own-label finished medicines via ANDA	Sun, Dr. Reddy's, Lupin, Zydus
Innovator Brand-owning drug company	Core	Core	Some	Core	Core	Monopoly pricing during the patent	Global: Novo Nordisk, Lilly

CDMO vs generic API maker. A CDMO works for an innovator during the patent, so client concentration and the project pipeline drive its earnings.
A generic API maker sells its own product after expiry, so ANDA approvals and price erosion drive its earnings. Some firms run both, so read segment disclosure.

Why contract revenue tends to stick, and why it is lumpy. Once a client's process is set up and filed with regulators, moving it to another plant means new filings, so clients rarely switch. But revenue follows the client's delivery schedule, so quarters can swing.

The same industry contains three business models with very different risk and reward

NCE, generics and biosimilars are a classification of what a company sells, not a list of new trends.

Exhibit 5 NCE, generics and biosimilars compared

MODEL	WHAT IT IS	RISK AND RETURN	WHERE THE MOAT IS	WHAT TO TRACK	INDIAN ANGLE
NCE New chemical entity	The innovator discovers a new molecule and patents it.	High risk, high reward	The patent, and clinical data no one else has	R&D as a % of revenue, pipeline stage, patent expiry calendar	Few listed Indian players; mostly global innovators
Generics	Copies of drugs whose patents have expired or been successfully challenged. Core of Indian pharma.	Lower risk, price competition, a volume game	Cost, scale, speed to file, clean plant record	ANDA pipeline and approval rate, US FDA status, price erosion	Indian firms supplied 47% of US generic prescriptions in 2022 (IQVIA)
Biosimilars	Versions of biologics that are "highly similar" to the original, with no clinically meaningful difference. Cannot be identical because they come from living cells.	Medium risk; needs capital and trials, but fewer rivals	Own cell lines and process, plus clinical evidence	Approvals, launch timing, interchangeability status, plant compliance	Biocon: biosimilars about 62% of group revenue in FY26

Source: FDA Biosimilar Basics; AAM 2026 Savings Report (with IQVIA); IQVIA Institute, U.S.-India Medicine Partnership (2024); Biocon FY26 results (biosimilars Rs 10,431 crore of Rs 16,927 crore). Risk and moat columns are the author's assessment.

Generics: volume, not value. In 2025, generics were 89% of US prescriptions (about 4 billion) but only 11.6% of spending (\$106 billion), against \$804 billion for brands. Generics save money, and the price competition squeezes margins.

Why biosimilars are harder. A small molecule can be copied exactly by chemistry. A biologic varies slightly from batch to batch even for its original maker, so a copier builds its own process and must prove the result matches. Biosimilar prices average roughly 40 to 50% below the original's price at launch (AAM), though discounts vary widely by product.

Oncology is the largest pool of drug spending and, with obesity, the fastest growing

A therapeutic area is the disease category a drug treats. It tells you who the customers are and how complex the molecules tend to be.

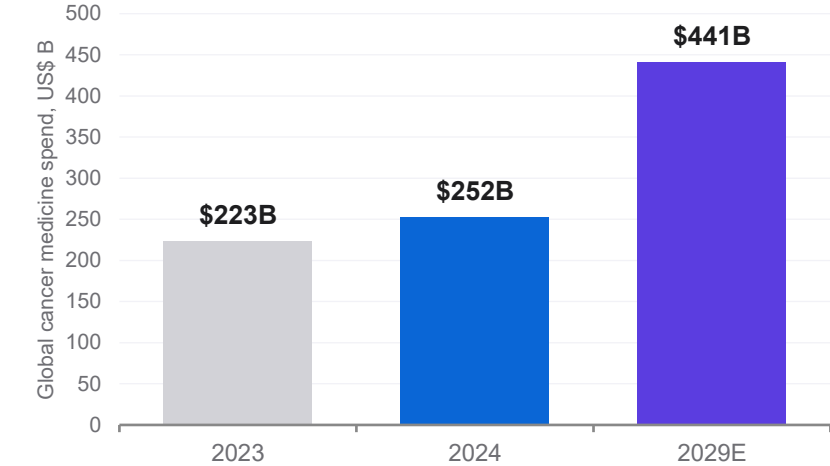
Exhibit 6 Therapeutic areas and what they mean for investors

AREA	WHAT IT COVERS	WHY IT MATTERS
Cardiovascular and metabolic	Heart disease, blood pressure, diabetes, obesity	Large chronic patient base. The GLP-1 class is the disruptor. Indian firms supplied 60% of US hypertension generic prescriptions in 2022.
Oncology	Treatment of cancer	Largest pool of drug spending. Molecules are complex (potent APIs, antibodies), so fewer firms can make them.
CNS	Central nervous system: brain and nerve conditions	Indian firms supplied 62% of US mental-health generic prescriptions in 2022, the highest share of any top category.
Anti-infectives	Antibiotics, antivirals	Price-competitive. Antibiotics are among the categories most dependent on imported inputs, mainly from China.
Rare disease and orphan drugs	Conditions with very few patients	High price per patient. The US exempted orphan and other specialty drugs from India from its new 100% tariff.

Source: IQVIA Institute, U.S.-India Medicine Partnership (2024); IQVIA Global Oncology Trends 2025; Rajya Sabha reply on API imports (FY25); US Commerce Department notice, 29 Sep 2026.

A note on labels. Chronic therapies (a treatment duration) and biologics (a drug type) cut across these areas. They are useful lenses, not therapeutic areas.

Exhibit 7 Global cancer medicine spend



Source: IQVIA Institute, Global Oncology Trends 2024 (2023) and 2025 (2024, 2029E), list-price basis.

Spend grew 75% over the five years to 2024. IQVIA ranks oncology as the single largest contributor to drug spending, ahead of immunology and obesity. Keytruda and Opdivo lose US patent protection around 2028, which is why biosimilars are a live theme.

Harder molecules mean fewer rivals and more pricing power, but cycles remain

This is the central idea for evaluating a pharma or CDMO business.

Exhibit 8 The complexity ladder

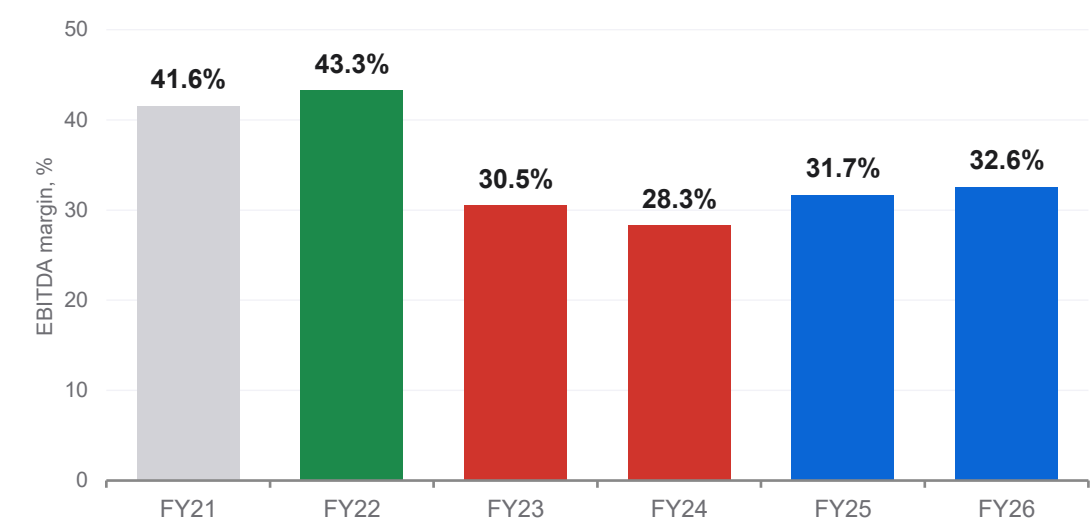
TIER	COMPETITION	PRICING POWER
1. Commodity generics and APIs	Many makers, including low-cost Chinese producers	Low. Price takers.
2. Complex generics, peptides, potent APIs	Fewer makers with the skills and plants	Moderate
3. Custom synthesis for innovators	Chosen supplier for a specific molecule, under contract	Higher, with longer contracts
4. Biologics and biosimilars	Few firms with cell-culture capacity and evidence	Highest, but heavy capital

Source: Author framework based on company and sector commentary; qualitative.

The barriers are physical as well as scientific. Highly potent oncology APIs are active at microgram doses, so plants need isolators, negative-pressure rooms and dedicated production lines. One such suite is quoted at \$20 to 30 million or more, and over 70% of this work is reported to be outsourced.

The suite cost and outsourcing share come from a CDMO's own marketing material, so treat them as indicative.

Exhibit 9 Divi's Laboratories EBITDA margin



What the margins show. In Q1 FY27, Divi's, where custom synthesis is about 60% of revenue, earned a 40.75% EBITDA margin. Laurus, which sells more generics, earned 31.8%. That gap is what pricing power looks like in practice. Margins still move with client demand and pricing: Divi's fell from 43.3% in FY22 to 28.3% in FY24, then recovered. Divi's is a broader custom synthesis and generics business rather than an oncology pure-play, so it illustrates the principle and does not prove it for oncology.

Indian firms compete across the chain, with China as both rival and supplier

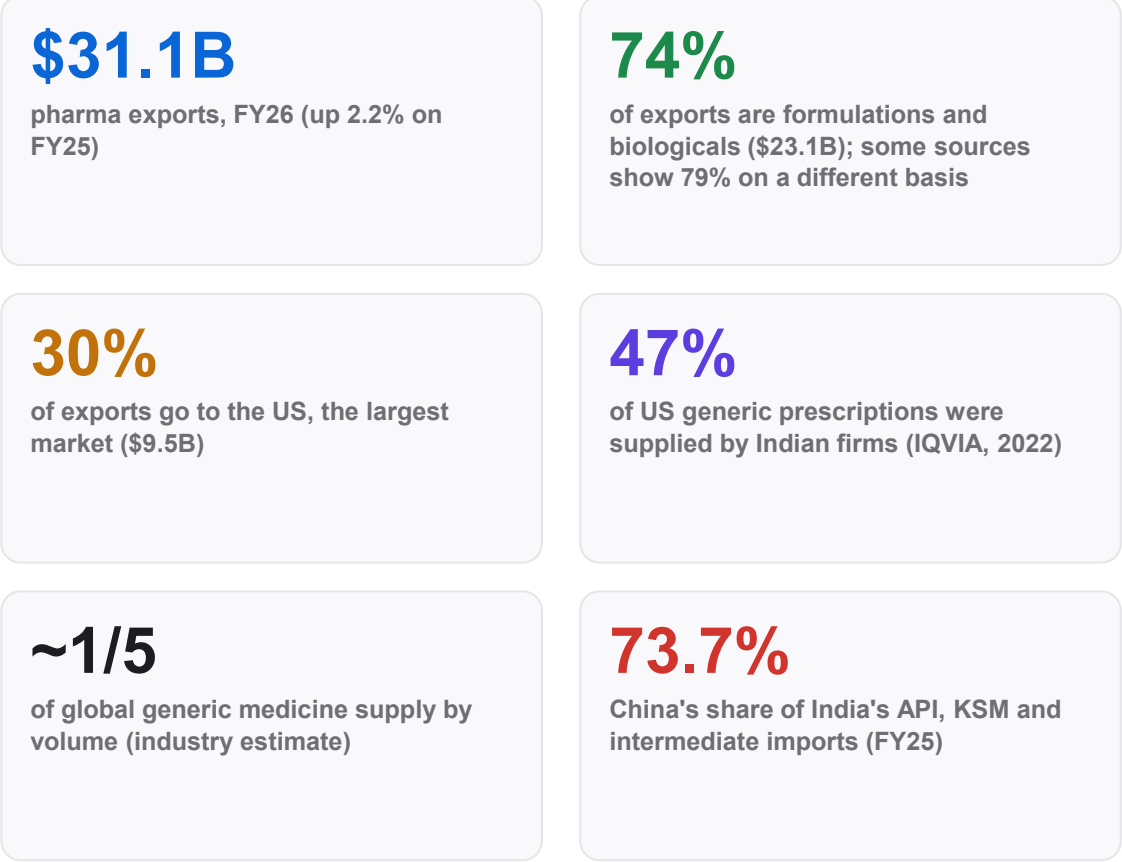
Who competes where, and how US policy on Chinese contract manufacturers is changing the picture.

Exhibit 10 Who competes where

LAYER	INDIAN PLAYERS (EXAMPLES)	MAIN GLOBAL COMPETITORS
Finished generics	Sun, Dr. Reddy's, Lupin, Zydus	Teva, Sandoz
Biosimilars	Biocon	Sandoz, Samsung Bioepis
Generic APIs	Divi's, Laurus	Chinese producers, which also supply most of India's raw materials
CDMO and CRDMO	Syngene, Sai Life, Divi's (custom synthesis), Cohance, Neuland	WuXi AppTec and WuXi Biologics (China), Lonza (Switzerland), Samsung Biologics (South Korea)

Source: Company disclosures; Jefferies and press coverage. Classification is simplified and some firms span several layers.

Exhibit 11 India by the numbers



Source: IBEF (DGCIS data); IQVIA Institute (2024); Rajya Sabha reply cited in BioSpectrum and Cargo Insights; Latham and Watkins, Ropes and Gray and Holland and Knight on BIOSECURE. KSM means key starting material, the raw ingredient an API is built from. India's share of US generic import value is estimated at about 30%.

China+1 in practice. The US BIOSECURE Act, signed on 18 December 2025, restricts federal contracts and grants that involve "biotechnology companies of concern". It does not name WuXi. But WuXi AppTec was added to the US defense department's list of Chinese military companies on 8 June 2026 and is challenging that in court, while WuXi Biologics is not listed. Implementing rules are still pending, so the effect on Indian CDMOs is a possibility, not yet a fact.

Why India is here. Until 2005, Indian law allowed only process patents on medicines, so firms built skill in making molecules by cheaper routes. That skill now sells as finished generics and as contract manufacturing.

Estimates of revenue losing exclusivity by 2030 range from \$175 billion to over \$300 billion

Evaluate puts it above \$300 billion globally; IQVIA about \$220 billion; Morgan Stanley about \$175 billion for the US alone. More of this wave is biologic than the last one.

Exhibit 12 Major expiries in the 2025 to 2030 wave

DRUG	OWNER	TYPE	2024 SALES	US EXPIRY	WHO COMPETES
Stelara	J&J	Biologic	\$10.4B	Open since 2025	Biosimilars (Biocon's Yesintek approved Dec 2024)
Eliquis	BMS, Pfizer	Small molecule	~\$13B (BMS)	Compound patent Nov 2026; settled generic entry 1 Apr 2028	Many generics
Keytruda	Merck	Biologic	\$29.5B	Late 2028	Biosimilars; Merck launched a subcutaneous version (Sep 2025)
Opdivo	BMS	Biologic	\$9.3B	Dec 2028	Biosimilars in development

Source: Company reports and press coverage of 2024 sales; Pfizer 10-K and BMS disclosures via DrugPatentWatch (Eliquis); court filings (Opdivo); Evaluate, IQVIA and Morgan Stanley estimates as reported by trade press.

A cliff is not automatic profit. ASPE found average US generic prices fell by half within a year of entry. The gains go to the first few entrants and the lowest-cost makers. Biosimilars also build share more slowly: one industry report puts the average at about 53% after five years. IQVIA expects biologics to be hit most in 2025 and 2026, and small molecules in 2027 and 2028.

Two ways to play the same cliff

- 1. Sell the generic or biosimilar.** Generic and biosimilar makers launch their own version. They win on approval speed, cost and plant compliance. The risk is price erosion.
- 2. Make the ingredient.** API suppliers and CDMOs supply whoever wins, and also innovators who need reliable capacity. They win on process chemistry and being qualified early. The risk is client concentration.

Why India is placed at both ends

Indian firms already hold US ANDA approvals and export volume, and a growing set of CDMOs is winning outsourcing as clients diversify away from China. Jefferies puts the China+1 opportunity for Indian CRDMOs at about \$700 million a year in its base case, and \$1.4 billion in a bull case.

What to look for. For generic and biosimilar makers: filings timed to the expiry date and a clean plant record. For suppliers: order books, capacity additions and how concentrated the client list is.

Semaglutide shows a patent cliff in real time, and it arrives country by country

Semaglutide is Novo Nordisk's molecule and one member of the GLP-1 class. Lilly's tirzepatide is a separate molecule that remains patent-protected.

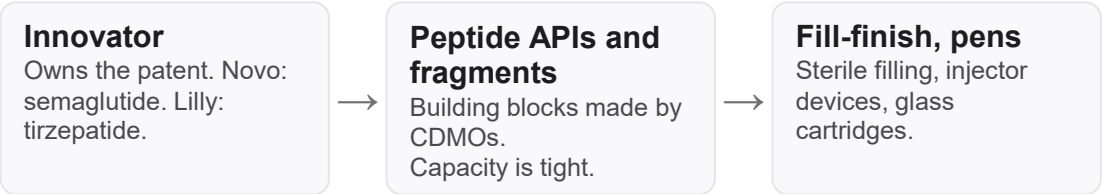
Exhibit 13 Where semaglutide's core patent stands

MARKET	CORE PATENT	WHAT HAPPENED
India	Expired 20 Mar 2026	Generics launched from 20 March; about 25 generic brands by April. Novo is litigating other patents, and a court said infringing product cannot be made, sold or exported.
Canada	Lapsed in 2019 (unpaid fee)	Dr. Reddy's got the first Health Canada approval (28 Apr); Apotex and Dr. Reddy's launched in mid-May. Novo reports lower prices, broadly stable volumes.
China, Brazil, Turkey	2026	Slower. In Brazil, no generic had launched as of Novo's Q1 report. A Chinese antitrust suit challenges Novo's extended exclusivity.
US, EU, Japan	Protected into the early 2030s	Apotex has a US tentative approval; Cipla filed a patent-challenge ANDA and Novo sued.

Source: Novo Nordisk 6-K filings (Q1 and H1 2026); Dr. Reddy's press releases; Health Canada; IQVIA (Jul 2025, Apr 2026); Pharmarack via BioSpectrum and Medical Dialogues.

Investor read-through: India's GLP-1 market reached Rs 1,736 crore in the year to April 2026, up from about Rs 545 crore a year earlier (Pharmarack). Lilly's tirzepatide was 63% of it. Generic semaglutide sold about Rs 44 crore in April alone, so generic semaglutide is still a small business.

Exhibit 14 The GLP-1 supply chain



Divi's Laboratories disclosed a long-term supply agreement for advanced intermediates with an unnamed global pharma company (May 2025), with Rs 650 to 750 crore of capacity funded by the customer's advance payments.

Jefferies believes it links to GLP drugs and estimates about \$630 million a year by FY32 from two intermediates for an oral small-molecule GLP drug, plus about \$426 million from peptide fragments for an injectable GLP drug. These are analyst estimates. Divi's has not named its customers.

Eli-Lilly plans more than \$1 billion of contract manufacturing in India and says it is engaging Indian contract manufacturers, without naming them.

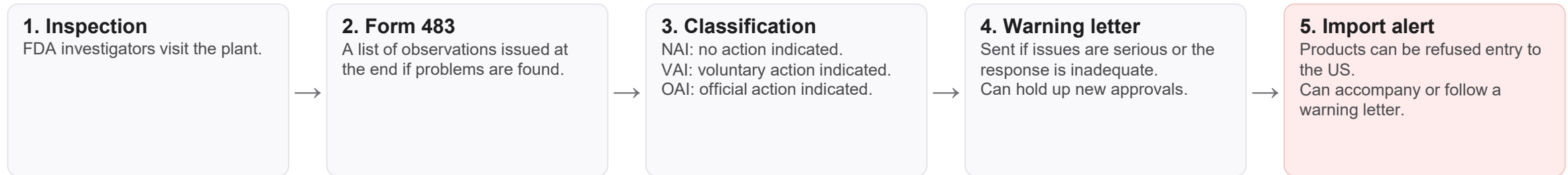
Bottleneck. The CEO of OneSource Specialty Pharma says demand exposes gaps across peptide APIs, sterile fill-finish, injector devices and glass cartridges, and expects constraints to last at least two years. That is one executive's view.

Source: Divi's exchange filings (May 2025); Jefferies via CB Insights; Reuters and Fierce Pharma on Lilly; Business Standard and Pharma Manufacturing on OneSource.

A plant's US FDA record is as important as its products

Check the status of every facility of every company you follow.

Exhibit 15 How an FDA inspection escalates



Source: US FDA inspection classification guidance and warning letter database.

● A real example

Dabur India's Silvassa plant (over-the-counter products) was inspected 12 to 16 January 2026 and received a Form 483. The FDA judged the company's 4 February reply inadequate, placed the site on Import Alert 66-40 on 5 June, and issued a warning letter on 24 July citing data integrity and quality-unit failures. Note that the import alert came first.

Source: FDA warning letter 320-26-105, 24 July 2026.

How to read it. A Form 483 alone is not a ban on US supply. What matters is the classification and whether the company's response closes the issues. FDA inspection classifications, warning letters and import alerts are public and searchable by facility.

● Other risks to keep on the list

- **China dependence.** China supplied 73.7% of India's API, KSM and intermediate imports in FY25 (68.2% in FY21), despite a Rs 6,940 crore incentive scheme running to FY30.
- **US tariffs (as of 29 Sep 2026).** A 100% default tariff on patented drugs and their ingredients applies from 31 July for 17 large companies and from 29 September for others. Generics and biosimilars are outside it for now, and Commerce must revisit generics within a year. India is among 20 jurisdictions with zero duty on specified specialty drugs. This is moving quickly.
- **Contract lumpiness.** CDMO quarters swing with client delivery schedules.

Eight things to check on any pharma or CDMO company, and why each one matters

Use it as a screen before reading a company's results.

Exhibit 16 What to track

#	METRIC	WHY IT MATTERS
1	ANDA pipeline and approval rate (US)	Shows the generic launches coming. The first company to challenge a patent can earn 180 days of exclusivity.
2	US FDA status of each facility	A plant on official-action status or import alert cannot ship to the US.
3	Regulated-market revenue mix	Revenue from the US, Europe and Japan faces tougher standards but generally holds better pricing.
4	R&D spend as a % of revenue (NCE)	Funds the high-risk bet. Too little means no pipeline; too much without a pipeline is cash burn.
5	API vs formulation split	Shows where the company sits in the chain and whether it is exposed to input or end-market pricing.
6	Client concentration (CDMO)	A few molecules can be most of revenue, so one client decision moves earnings.
7	KSM sourcing geography	Heavy China sourcing raises supply and geopolitical risk; backward integration lowers it.
8	Patent expiry calendar	Shows revenue at risk for innovators and opportunity windows for copiers.

● The closing idea

Moving from commodity generics to complex molecules and biologics, manufacturing gets harder, fewer companies can do it, and those that can hold more pricing power. Complex oncology APIs and biosimilars are where that shows up most clearly.

● Signs a company sits higher on the ladder

- Revenue comes from multi-year contracts, not spot generic sales.
- Few rivals can make its key products.
- Its plants are approved for complex work: potent APIs, peptides, sterile injectables or biologics.
- Its EBITDA margin sits well above generic-heavy peers.